

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**AI and application Development and jira**

**Practical – 7- D& E**

|                              |                                   |
|------------------------------|-----------------------------------|
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| Class:TYIT                   | Batch:1                           |
| Date of Assignment:07-8-2026 | Date/Time of Submission:07-8-2026 |

**Aim:-**7D. Implement k-fold cross-validation on a classification model.

**Code:-Give dataset**

```
from sklearn.datasets import load_iris
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
# Load dataset
iris = load_iris()
X = iris.data
y = iris.target
# Create classification model
model = KNeighborsClassifier()
# Apply 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5)
print("Accuracy scores for each fold:")
print(scores)
print("Average Accuracy:", round(scores.mean() * 100, 2),
"%")
print("Aashu 50")
```

**Output:-**

```
Accuracy scores for each fold:
[0.96666667 1.         0.93333333 0.96666667 1.         ]
Average Accuracy: 97.33 %
Aashu 50
```

| Iris Dataset                | Wine Dataset                                 |
|-----------------------------|----------------------------------------------|
| 150 samples, 4 features     | 178 samples, 13 features                     |
| Accuracy: <b>96–98%</b>     | Accuracy: <b>65–75%</b><br>(approx.)         |
| Simple and easy to classify | More complex dataset                         |
| Better performance          | Lower performance<br>without feature scaling |

**Code:-**

**Different dataset**

```
from sklearn.datasets import load_wine
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
# Load dataset
wine = load_wine()
X = wine.data
y = wine.target
# Create classification model
model = KNeighborsClassifier(n_neighbors=3)
# Apply 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5)
print("Accuracy scores for each fold:")
print(scores)
print("Average Accuracy:", round(scores.mean() * 100, 2),
"%")
print("Aashu 50")
```

**Output:-**

```
Accuracy scores for each fold:
[0.63888889 0.69444444 0.66666667 0.65714286 0.85714286]
Average Accuracy: 70.29 %
Aashu 50
```

**Justification:** The **Iris dataset** performs better than the **Wine dataset** because it is simpler and has fewer features,

## Aashu Yadav

### T050

resulting in higher classification accuracy.

**Conclusion:** Iris gives higher accuracy due to its simplicity, while Wine is more complex and needs feature scaling for better KNN performance.

### 7E. Implement a Decision Tree classifier using the Iris or any relevant dataset.

#### Code:-

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
iris = load_iris()
X = iris.data
y = iris.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1)
# Create model
model = DecisionTreeClassifier()
# Train model
model.fit(X_train, y_train)
# Display accuracy
print("Training Accuracy:", model.score(X_train, y_train)
* 100)
print("Testing Accuracy:", model.score(X_test, y_test) *
100)
prediction = model.predict([X_test[0]])
print("Predicted Class:", prediction[0])print("Aashu 50")
```

#### output:-

```
Training Accuracy: 100.0
Testing Accuracy: 95.55555555555556
Predicted Class: 0
Aashu 50
```

#### Code:-

```
from sklearn.datasets import load_breast_cancer
```

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
data = load_breast_cancer()
X = data.data
y = data.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1)
# Create model
model = DecisionTreeClassifier()
# Train model
model.fit(X_train, y_train)
# Display accuracy
print("Training Accuracy:", model.score(X_train,
y_train)*100)
print("Testing Accuracy:", model.score(X_test, y_test) *
100)prediction = model.predict([X_test[0]])
print("Predicted Class:", prediction[0])print("Aashu 50")
```

#### Output:-

```
Training Accuracy: 100.0
Testing Accuracy: 94.73684210526315
Predicted Class: 1
Aashu 50
```

**Justification:** The Breast Cancer dataset is more complex with more features, while the Iris dataset is simpler and easier for the Decision Tree classifier to learn.

| Iris Dataset                       | Breast Cancer Dataset                |
|------------------------------------|--------------------------------------|
| 150 samples, 4 features            | 569 samples, 30 features             |
| 3 classes                          | 2 classes                            |
| Higher accuracy due to simple data | More complex but gives good accuracy |
| Faster training                    | Requires more computation            |